CITY UNIVERSITY MALAYSIA

Est. 1984

BIT1063 : Discrete Mathematics

202605 | MAY 2026

ASSIGNMENT (Group)

No	Name	Student ID	Program
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SUBMISSION DUE: WEEK 13

SUBMITTED TO: MR MOHD NOOR HAFIZEE BIN YAHAYA

AUTHENTICATION & DECLARATION OF ORIGINALITY

We hereby declare that this assignment is our own original work and has been prepared specifically for this course. All sources of information, ideas, and materials from other works have been properly cited and acknowledged in accordance with academic standards. We confirm that this assignment has not been submitted, either in whole or in part, for assessment in any other course or institution.

We further acknowledge that any form of plagiarism, fabrication, or falsification constitutes academic misconduct and may result in disciplinary action in accordance with faculty regulations.

By signing below, each group member confirms their individual contribution to the assigned section and accepts joint responsibility for the integrity and accuracy of the entire submission.

Group Member Authentication

No.	Student Name	Student ID	Assigned Section	Signature	Date
1					
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Submission Acknowledgement (For Office Use)

Received By	Date	Signature

Important Notes for Students

• This page must be printed and signed by all group members.

• Unsigned or partially signed declarations may result in non-acceptance of the submission.

• Digital signatures are not permitted unless officially approved.

COURSE LEARNING OUTCOME (CLO)

Analyze computational problems using mathematical reasoning (C4, PLO7).

ASSIGNMENT PURPOSE

This assignment is designed to assess students' ability to apply discrete mathematical concepts to solve complex computational problems. Students are required to demonstrate analytical thinking, formal reasoning, and structured problem solving, rather than descriptive explanation.

IMPORTANT INSTRUCTIONS

1. Each student must complete only their assigned section.
2. Mathematical solutions must be handwritten.
3. Explanations, interpretations, and analysis must be typed.
4. The final report must be printed and submitted physically, together with a softcopy.
5. Failure to follow format requirements may result in mark deduction.

SUBMISSION REQUIREMENTS

1. Handwritten Submission (Compulsory)

The following must be handwritten and compiled under Appendix A:

- Definitions of symbols and variables
- Logical expressions and proofs
- Truth tables
- Mathematical derivations
- Set diagrams, trees, and graphs
- Step-by-step calculations

2. Typed Report (Compulsory)

The following must be typed:

- Problem interpretation
- Mathematical reasoning and justification
- System analysis and discussion
- Conclusions and reflections

The typed report must be:

- Printed and submitted in hardcopy
- Submitted in PDF format (softcopy)

INDIVIDUAL CASE STUDIES AND TASKS

SECTION 1: PROPOSITIONAL LOGIC CASE STUDY

Smart Library Access System

A university smart library uses an automated gate system to determine whether a user may enter a restricted study area. Entry depends on whether the student has an active library account, whether their student card is valid, whether the study room has available seats, and whether a librarian grants temporary access.

Problem-Solving Tasks

Handwritten

1. Define propositional variables for all conditions.
2. Construct a compound logical expression representing the access rule.
3. Create a complete truth table.
4. Determine whether unauthorized access is possible under any condition.

Typed Report

1. Explain the logical behavior of the access system.
2. Identify any redundant propositions.
3. Simplify the logical expression using logical equivalence laws.
4. Discuss how logical errors could affect library security.

SECTION 2: PREDICATE LOGIC AND QUANTIFIERS CASE STUDY

Scholarship Eligibility System

A university scholarship is awarded only to students who satisfy all academic requirements, submit all required documents, and maintain a minimum CGPA. However, some eligible students have been rejected while others were mistakenly approved.

Problem-Solving Tasks

Handwritten

1. Define appropriate predicates, variables, and domains.
2. Express the scholarship rules using quantified statements.

3. Negate one quantified rule formally.
4. Use the negation to illustrate a possible system error.

Typed Report

1. Explain the importance of quantifier placement.
2. Analyze how incorrect quantifiers may affect eligibility decisions.
3. Propose a corrected logical formulation.
4. Justify the formulation mathematically.

SECTION 3: SET THEORY CASE STUDY

University Club Membership Management

Students may belong to various clubs such as the Robotics Club, Debate Club, Volunteer Society, and Sports Club. Some students join multiple clubs, while others join none. The Student Affairs Office wants to analyze membership overlaps to improve event planning.

Problem-Solving Tasks

Handwritten

1. Define the universal set and relevant subsets.
2. Apply union, intersection, complement, and difference operations.
3. Determine overlapping memberships.
4. Draw a Venn diagram illustrating the relationships.

Typed Report

1. Interpret the meaning of each set operation.
2. Discuss how overlapping memberships affect event scheduling.
3. Analyze issues caused by poorly defined sets.
4. Justify the chosen set structure.

SECTION 4: SEQUENCES CASE STUDY

Campus Wi-Fi User Growth

The number of devices connected to the campus Wi-Fi network increases each week. Initially, growth appears constant, but later observations suggest a rapid increase due to new student enrollment.

Problem-Solving Tasks

Handwritten

1. Model the growth using an arithmetic sequence.
2. Model the growth using a geometric sequence.
3. Derive the nth-term formula for both models.
4. Predict when the network limit will be exceeded.

Typed Report

1. Compare arithmetic and geometric growth patterns.
2. Determine the most suitable model.
3. Justify your selection using calculations.
4. Discuss implications for network infrastructure planning.

SECTION 5: SERIES CASE STUDY

Cloud Storage Subscription Costs

A university department subscribes to cloud storage services. Subscription costs increase annually due to storage expansion and service upgrades. Management wants to estimate the total expenditure over several years.

Problem-Solving Tasks

Handwritten

1. Model the costs using arithmetic and geometric series.
2. Calculate the total expenditure over a specified period.
3. Show all derivation steps.

Typed Report

1. Interpret the cumulative costs.
2. Compare long-term impacts of both models.

3. Evaluate financial sustainability.
4. Discuss budgeting risks.

SECTION 6: RELATIONS CASE STUDY

Lecturer–Course Assignment System

A university maintains a system that assigns lecturers to courses. Some lecturers teach multiple courses, while some courses are taught by multiple lecturers.

Problem-Solving Tasks

Handwritten

1. Define the relation using ordered pairs.
2. Determine the domain and range.
3. Classify the relation type.
4. Represent the relation using a table or matrix.

Typed Report

1. Analyze potential scheduling conflicts.
2. Discuss limitations of the assignment structure.
3. Explain the significance of the relation type.
4. Recommend improvements to assignment management.

SECTION 7: FUNCTIONS CASE STUDY

Student Grade Conversion System

A university system converts raw assessment scores into final percentage grades using a mathematical formula. Administrators want to know whether original scores can always be recovered from the final grades.

Problem-Solving Tasks

Handwritten

1. Define the grading function.
2. Determine whether the function is one-to-one.
3. Find the inverse function if it exists.
4. Verify the inverse mathematically.

Typed Report

1. Explain the meaning of invertibility.
2. Analyze whether information is lost during conversion.
3. Discuss limitations of the grading model.
4. Suggest possible improvements.

SECTION 8: TREES CASE STUDY

Online Course Recommendation System

An online learning platform recommends courses through a series of decision points based on student interests, skill levels, and academic background.

Problem-Solving Tasks

Handwritten

1. Construct a rooted decision tree.
2. Perform preorder traversal.
3. Perform inorder traversal.
4. Perform postorder traversal.

Typed Report

1. Analyze the efficiency of the recommendation process.
2. Identify unnecessary decision branches.
3. Suggest improvements to reduce complexity.
4. Justify recommendations using tree properties.

SECTION 9: GRAPH THEORY (REPRESENTATION) CASE STUDY

Shuttle Bus Route Optimization

A university shuttle service operates between campus locations such as dormitories, lecture halls, cafeterias, and libraries. Management wants to optimize transportation efficiency.

Problem-Solving Tasks

Handwritten

1. Represent the locations as a weighted graph.
2. Identify meaningful subgraphs.

3. Compute shortest routes between selected locations.
4. Illustrate the graph clearly.

Typed Report

1. Interpret the shortest path results.
2. Analyze route efficiency.
3. Evaluate strengths of the graph model.
4. Discuss practical limitations.

SECTION 10: GRAPH THEORY (CONNECTIVITY AND PROPERTIES) CASE STUDY

Campus Maintenance Inspection Routes

Maintenance staff must inspect buildings connected by walkways across campus. Administrators want to know whether all pathways can be inspected without retracing routes and whether every building can be visited exactly once.

Problem-Solving Tasks

Handwritten

1. Analyze graph connectivity.
2. Determine whether Eulerian paths or circuits exist.
3. Determine whether Hamiltonian paths or circuits exist.
4. Provide mathematical justification.

Typed Report

1. Interpret the results in practical terms.
2. Analyze operational efficiency.
3. Evaluate network robustness.
4. Recommend infrastructure improvements based on graph properties.

FINAL REPORT STRUCTURE (MANDATORY)

Section No.	Title	Recommended Pages
1	Cover Page	1
2	Authentication	2
3	Introduction	1 - 2
4	Case Study 1: Propositional Logic	2 - 3
5	Case Study 2: Predicate Logic and Quantifiers	2 - 3
6	Case Study 3: Set Theory	2 - 3
7	Case Study 4: Sequences	2 - 3
8	Case Study 5: Series	2 - 3
9	Case Study 6: Relations	2 - 3
10	Case Study 7: Functions	2 - 3
11	Case Study 8: Trees	2 - 3
12	Case Study 9: Graph Theory - Representation	2 - 3
13	Case Study 10: Graph Theory - Connectivity and Properties	2 - 3
14	Integrated System Analysis	2
15	Discussion	1 - 2
16	Conclusion	1
17	Group Reflection	1
18	References	1
19	Appendix A: Handwritten Mathematical Solutions	Not Included in Page Count

ASSESSMENT RUBRICS

Rubric 1: Individual Contribution (70%)

Total Individual Marks:
8 criteria × 4 marks = 32 marks, scaled to 70%

Criteria	Excellent (4 marks)	Good (3 marks)	Satisfactory (2 marks)	Weak / Unsatisfactory (1 mark)
Problem Interpretation	Demonstrates complete and accurate understanding of the case study. All assumptions are clearly stated and logically justified.	Shows good understanding of the problem with minor unclear assumptions.	Basic understanding; some assumptions are missing or weak.	Misinterprets the problem or assumptions are incorrect.
Mathematical Modelling	The mathematical model is correct, appropriate, and well-structured. All variables and symbols are clearly defined.	The model is mostly correct with minor structural or notation issues.	The model partially represents the problem but contains gaps or inaccuracies.	The model is incorrect or does not represent the problem.
Logical Reasoning & Analysis (C4)	Demonstrates strong analytical reasoning. Conclusions are logically derived and fully justified.	Reasoning is generally sound but lacks depth in some areas.	Limited analysis; conclusions are weakly supported.	No real analysis; mostly descriptive or incorrect reasoning.

Handwritten Work Quality	All handwritten work is complete, neat, and easy to follow. Steps are logically ordered.	Handwritten work is mostly clear with minor omissions.	Handwritten work is difficult to follow or incomplete in places.	Handwritten work is unclear, disorganised, or largely missing.
Accuracy of Calculations / Proofs	All calculations, proofs, and derivations are correct.	Minor errors that do not affect overall conclusions.	Multiple errors affecting reliability of results.	Calculations or proofs are mostly incorrect or absent.
Use of Mathematical Notation	Notation is precise, consistent, and used correctly throughout.	Mostly correct notation with minor inconsistencies.	Inconsistent or partially incorrect notation.	Incorrect or inappropriate notation.
Interpretation of Results	Results are interpreted correctly and linked clearly to the real-world case.	Interpretation is mostly correct but lacks depth.	Interpretation is superficial or partially incorrect.	Results are not interpreted or are misinterpreted.
Clarity of Written Explanation	Typed explanation is clear, coherent, and well-organised.	The explanation is understandable but could be more structured.	The explanation is unclear or poorly organised.	The explanation is confusing or largely missing.

Rubric 2: Group Integration and Collaboration (30%)

Total Group Marks:

7 criteria × 4 marks = 28 marks, scaled to 30%

Criteria	Excellent (4 marks)	Good (3 marks)	Satisfactory (2 marks)	Weak / Unsatisfactory (1 mark)
Conceptual Integration	All individual sections are strongly connected and form a single coherent system analysis.	Most sections are connected with minor inconsistencies.	Sections are loosely connected with limited integration.	Sections are isolated with no clear integration.
Consistency of Assumptions	Assumptions are consistent across all sections. No contradictions present.	Minor inconsistencies that do not affect overall analysis.	Multiple inconsistencies across sections.	Assumptions contradict each other.
Cross-Referencing Between Sections	Sections explicitly reference and support each other logically.	Some cross-referencing presents.	Minimal cross-referencing.	No cross-referencing at all.
Balance of Contribution	Contributions are well balanced and responsibilities are clearly fulfilled.	Slight imbalance but all members contribute meaningfully.	Noticeable imbalance in contribution.	One or more members contribute very little.

Collaboration Evidence	Strong evidence of collaboration (shared models, consistent notation, unified discussion).	Evidence of collaboration with minor gaps.	Limited evidence of teamwork.	No clear evidence of collaboration.
Overall Report Organisation	The report is professionally structured and easy to follow.	Organisation is generally clear with minor issues.	Organisation is weak and confusing in places.	The report is poorly organised.
Presentation Quality (Printed Report)	The printed report is neat, professional, and error-free.	Minor formatting or language issues.	Multiple formatting or language errors.	Poor presentation quality.

FINAL NOTES

- Academic Integrity. All submitted work must be original. Plagiarism, including copying from other groups, reuse of senior work, or unauthorised use of generative tools without understanding, constitutes academic misconduct and will be handled in accordance with faculty disciplinary regulations.
- Individual Accountability. Although this is a group assignment, each student is assessed individually based on their assigned case study. Failure to complete an assigned section satisfactorily may significantly affect the individual mark regardless of overall group performance.
- Group Responsibility. The group component evaluates the coherence and integration of all sections. Weak integration, inconsistent assumptions, or lack of collaboration will negatively affect the group marks awarded to all members.
- Submission Format Compliance. All mathematical derivations, proofs, diagrams, and calculations must be handwritten and included in the appendix. Analytical explanations, interpretations, and discussions must be typed. Failure to follow the required format may result in mark deduction.
- Printed Submission Requirement. A printed hardcopy of the typed report is compulsory. Submissions without a printed report, or reports that are incomplete or poorly presented, may be penalised.
- Clarity and Legibility. Handwritten work must be clear, well-organised, and legible. Illegible handwriting, missing steps, or poorly scanned pages may be treated as incomplete work.
- Late Submission. Late submissions are subject to penalties as specified by faculty policy unless valid supporting documentation is provided and approved in advance.
- Marking Transparency. Assessment will be conducted strictly according to the published rubrics. Students are strongly advised to review the rubric carefully and ensure that their work explicitly meets the stated criteria.
- Professional Standard. The final report must reflect professional academic standards, including proper mathematical notation, structured argumentation, and formal language throughout.